
Refinement Steps Buy Intelligibility, Not Identity: Metric-Selective Test-Time Scaling in Masked-Diffusion TTS

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 20–125 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 45 trained models. Fitting $\text{err} =$
7 $E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ pre-registered, refinement steps attack intelligibility
8 harder than speaker identity: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$, 95% run-level bootstrap
9 CI [0.092, 0.131]. Word error rate saturates only at $T^*=16$ while identity saturates
10 at 8. Depth and steps are *not* interchangeable: the substitution form $B(dT^\kappa)^{-\beta}$
11 is worse than the separable form by $\Delta\text{AICc} = +69.3$. Steps are an intelligibility
12 knob, not a general one.

1 Introduction

14 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
15 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
16 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
17 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
18 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
19 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

20 Zero-shot text-to-speech is an unusually clean probe of that question, because it separates two
21 capabilities that are measured by standard, independent metrics. Intelligibility is compositional: the
22 model must place the right phonetic content in the right order, and word error rate (WER) reads it
23 out. Speaker identity is representational: the model must carry timbre from a short prompt into new
24 content, and speaker similarity (SIM-o) reads that out. The two need not respond to the same knob.

25 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
26 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
27 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve
28 intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility,
29 steps and depth exchange at a measurable rate κ — how many step-doublings buy one depth-doubling.

$$\text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \quad \text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \quad (1)$$

30 Both claims were registered before the first run, together with the thresholds, the frozen sampler
31 and the analysis plan. The answer to the first is yes and quantitative: $\Delta\tau = 0.110$ with a run-level

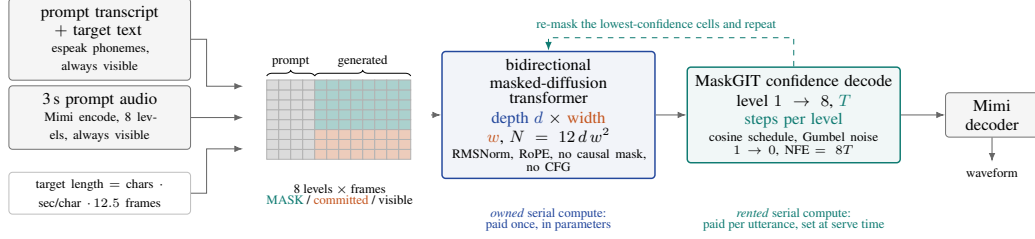


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully masked. The backbone spends *depth* — *owned* serial compute, fixed at training time — and the sampler spends *T refinement steps per codebook level* — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

bootstrap CI of $[0.092, 0.131]$, so the step exponent is reliably larger for intelligibility than for identity. The answer to the second is a clean no: the substitution form is decisively *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o) and κ runs to its bound, so no finite depth-step exchange rate describes this regime. The two results together say something sharper than either alone: extra inference compute is a targeted instrument, and it cannot be treated as a substitute for architecture.

- ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from one training grid of 45 models, released with the harness.
- ❷ $\Delta\tau = 0.110$, CI $[0.092, 0.131]$ — steps are metric-selective — and the *negative* result on depth-step substitution stated as an AICc comparison ($+69.3$) rather than a narrative, with κ reported as unidentified.
- ❸ A serving rule measured rather than assumed: c_{layer} is flat in *width* at batch 1 on the evaluation GPU, so serial latency is $8Td c_{\text{layer}}$ and the *step budget* is the only latency knob that trades against intelligibility.

2 Related work

Masked diffusion language models are now trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-capability question cannot arise. In speech, the masked-generative recipes that made codec-token TTS practical — confidence-decoded parallel generation over residual codebook levels [Borsos et al., 2023, Chang et al., 2022] and its multi-stage successors [Wang et al., 2024] — fixed a step budget by engineering judgement and reported a single operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has begun to report scaling behaviour, but in total parameters N only [Ye et al., 2025], which cannot separate width from depth and has no step axis at all; classical scaling work is likewise N -indexed [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is measured on one correctness metric; whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech. That is the gap this abstract fills: per-metric laws in (w, d, T) for a masked-diffusion TTS model.

3 Method

Backbone and objective. A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus MASK and PAD). *Width* w , *depth* d , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE, no weight tying, so the non-embedding parameter count is $N = 12dw^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training is SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , levels above l are entirely MASK; the loss is cross-entropy on the masked cells of level l only. A phoneme

prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so the backbone itself is the only identity pathway, and no classifier-free guidance, which would add a second inference-time axis and confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

Frozen sampler. MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with exactly T refinement steps per level, so $\text{NFE} = 8T$: a cosine schedule sets how many cells remain masked after each step, tokens are drawn categorically at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$ within a level; committed cells stay committed. The sampler is identical for every config, seed and T — nothing is tuned per shape or per step budget. Per-item generation noise is keyed so that each item sees the same noise at the first step of every level for every T , making T the only difference between surface points of one run. Before any metric is computed we verify that T reaches the sampler: outputs at $T=1$ and $T=16$ must differ on more than 20% of generated token cells.

Design and statistics. 15 configs span three iso- N budgets $\times$ five aspect ratios, each trained at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256, base LR 0.004); width transfer and depth transfer were verified by five-point LR sweeps before the grid. $T \in \{1, 2, 4, 8, 16\}$ is swept at evaluation only, giving 75 surface points per metric from 45 trained models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy, the Whisper English normaliser), SIM-o (WavLM-large [Chen et al., 2022] speaker-verification embedding against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022] as a tertiary. Error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. Both forms of Eq. (1) are fitted by weighted non-linear least squares with 32 multi-starts; $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ , and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ carry percentile confidence intervals from a bootstrap that resamples at the *run* level — the five T -points of one run share a trained model, so resampling surface points would understate their correlation — with 2000 replicates recomputing every T point.

4 Results

Steps are metric-selective (H-D2, supported). Fitted on all 75 surface points, the step exponent is $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), so $\Delta\tau = 0.110$ with CI [0.092, 0.131], excluding zero. The raw surface shows it without any model: over the grid, $T=1 \rightarrow T=16$ takes WER from 113.8% to 18.7% (a factor of $6.1\times$ on the error scale) while SIM-o moves only $0.209 \rightarrow 0.370$ ($1.25\times$ on $1 - \text{SIM-o}$), and the saturation steps differ: $T_{\text{WER}}^* = 16$ against $T_{\text{SIM}}^* = 8$. The power gate passed in all three budgets before these tests were run.

Depth and steps do not substitute (H-D3, not supported). The pre-registered substitution form collapses: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the hypothesis required $\leq +4$, and the fitted κ sits on its bound for both metrics, i.e. the exchange rate is not identified. We therefore quote no κ . The mechanism is visible in the fits: T enters with a large additive term that decays as $T^{-\tau}$ towards a floor, whereas depth changes the floor itself; a model that forces the two into one factor dT^κ cannot fit both the strong depth ordering at fixed T and the strong T ordering at fixed depth.

Shape matters, but the anisotropy ratio is not identified (H-D1, not supported). At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AICc} = -15.5$ for WER, -5.5 for SIM-o, against a -4 threshold), so error is not a function of parameter count alone — iso- N shape choice matters. The direction of the anisotropy, however, we cannot claim: $\Delta\rho = 0.174$ with CI [-0.711, 0.388], spanning zero, and the width coefficient A collapses to its lower bound for both metrics at $T=16$, which leaves α — and hence $\rho = \alpha/\beta$ — unidentified. What the data do show unambiguously is a strong *depth* ordering at fixed parameter count: moving from the shallowest shape to the best-performing depth is worth 40.2, 16.3 and 7.3 WER points in the 20, 50 and 125 M budgets, while SIM-o varies by under 0.065 across all 15 shapes. The gain saturates rather than continuing: the optimum is *interior* in the two smaller budgets ($d = 18$ at 4–24; $d = 30$ at 6–30; $d = 36$ at 8–36), so the design rule is "deep enough", not "as deep as possible".

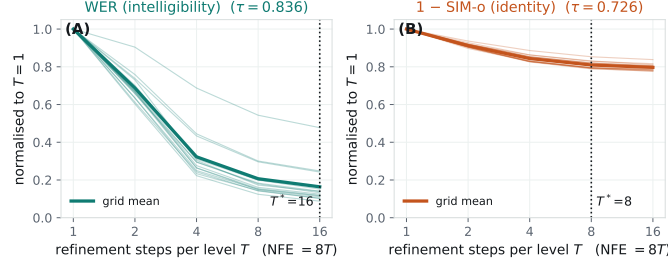


Figure 2: *Test-time scaling is metric-selective.* Error versus **refinement steps** for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means and the dotted line marks the pooled saturation step T^* . **(A)** WER is still falling at $T=16$ ($T_{\text{WER}}^* = 16$). **(B)** SIM-o has flattened by $T=8$. Same models, same frozen sampler, same items — only T differs.

Table 1: *Fitted exponents on the 75-point surface* (M_{sep} for α, β, τ ; M_{sub} for κ) with 95% run-level bootstrap CIs for the step exponents. $\Delta\tau = 0.110$, CI [0.092, 0.131], excludes zero: the pre-registered primary result. κ is at its bound in both rows and $\Delta\text{AIC}_{\text{sub-sep}} \gg +4$, so no exchange rate is quotable. $\Delta\text{AIC}_{\text{full-N}} \leq -4$ means iso- N shape matters.

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{sub-sep}}$	$\Delta\text{AIC}_{\text{full-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

Extrapolation and reliability. Fitting the two smaller budgets and predicting the largest gives MAPE 2.4% for SIM-o (versus 2.2% for the N -only model), inside the 15% the hypothesis asked for; WER does not extrapolate (20.2% versus 20.4%). Degenerate outputs — empty, truncated or looping transcripts under the frozen rule — fall monotonically with the step budget, from 40.4% at $T=1$ to 0.1% at $T=16$: most of what $T=1$ costs is not subtle quality but outright failures.

Serving corollary. Measured on the evaluation GPU at batch 1, c_{layer} is flat in width across the whole grid (0.493–0.514 ms per layer for $w \in [256, 1152]$): at these sizes the device is launch-bound, so serial latency is $8Td c_{\text{layer}}$ and width is free. The allocation rule follows: spend parameters on **depth** until latency binds, then spend the remaining latency on **steps**, which buy intelligibility to $T_{\text{WER}}^* = 16$ but stop buying identity after 8. A system that needs identity more than intelligibility can run at 8 steps and reclaim 50% of a $T=16$ configuration’s serial latency.

5 Limitations and outlook

The ceiling is 133 M non-embedding parameters and 30,000 steps; every model is under-trained by frontier standards, and the T curves may flatten differently at scale — $T_{\text{WER}}^* = 16$ is a lower bound, since WER had not saturated at the largest budget we measured. One language, one codec, one corpus, and no classifier-free guidance — guidance is a second inference-time axis that would confound the step axis, so excluding it is deliberate, and it is the obvious follow-up. The $\Delta\tau$ interval resamples 3 runs per config, so it captures seed-to-seed variation only, not the uncertainty of the single shared harness and corpus. $\Delta\rho$ and κ are reported as unidentified rather than estimated. Cuts taken: none. This is pre-registered outcome class S1, and selectivity is the only positive claim the data license.

Reproducibility. Protocol, hypotheses, thresholds, frozen sampler and analysis plan were registered before the first run; every gate is logged with its measured values. The released artefact holds the harness, all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

References

- Philip Anastassiou et al. Seed-TTS: A family of high-quality versatile speech generation models. *arXiv preprint arXiv:2406.02430*, 2024.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Zalán Borsos, Matt Sharifi, Damien Vincent, Eugene Kharitonov, Neil Zeghidour, and Marco Tagliasacchi. SoundStorm: Efficient parallel audio generation. *arXiv preprint arXiv:2305.09636*, 2023.
- Huiwen Chang, Han Zhang, Lu Jiang, Ce Liu, and William T. Freeman. MaskGIT: Masked generative image transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Michael Zeng, Xiangzhan Yu, and Furu Wei. WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 2022.
- Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*, 2025.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning (ICML)*, 2023.
- Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi Saruwatari. UTMOS: UTokyo-SaruLab system for VoiceMOS challenge 2022. In *Interspeech*, 2022.
- Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- Chengyi Wang, Sanyuan Chen, Yu Wu, Ziqiang Zhang, Long Zhou, Shujie Liu, Zhuo Chen, Yanqing Liu, Huaming Wang, Jinyu Li, Lei He, Sheng Zhao, and Furu Wei. Neural codec language models are zero-shot text to speech synthesizers. *arXiv preprint arXiv:2301.02111*, 2023.
- Yuancheng Wang, Haoyue Zhan, Liwei Liu, Ruihong Zeng, Haotian Guo, Jiachen Zheng, Qiang Zhang, Xueyao Zhang, Shunsi Zhang, and Zhizheng Wu. MaskGCT: Zero-shot text-to-speech with masked generative codec transformer. *arXiv preprint arXiv:2409.00750*, 2024.

191 Greg Yang, Edward J. Hu, Igor Babuschkin, Szymon Sidor, Xiaodong Liu, David Farhi, Nick
 192 Ryder, Jakub Pachocki, Weizhu Chen, and Jianfeng Gao. Tensor programs V: Tuning large neural
 193 networks via zero-shot hyperparameter transfer. In *Advances in Neural Information Processing*
 194 *Systems (NeurIPS)*, 2021.

195 Zhen Ye, Xinfu Zhu, Chi-Min Chan, Xinsheng Wang, Xu Tan, Jiahe Lei, Yi Peng, Haohe Liu, Yizhu
 196 Jin, Zheqi Dai, Hongzhan Lin, Jianyi Chen, Xingjian Du, Liumeng Xue, Yunlin Chen, Zhifei Li,
 197 Lei Xie, Qiuqiang Kong, Yike Guo, and Wei Xue. Llasa: Scaling train-time and inference-time
 198 compute for LLaMA-based speech synthesis. *arXiv preprint arXiv:2502.04128*, 2025.

199 A The shape grid, as run

200 Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each).
 201 Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the
 202 five shapes per budget span the aspect-ratio range at fixed $N = 12dw^2$.

config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

204 B Additional figures

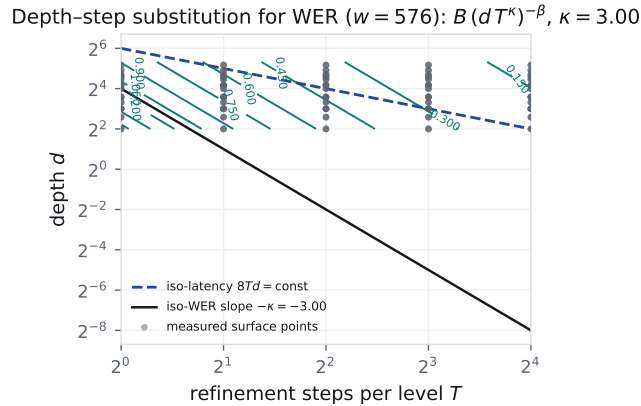


Figure 3: *Why the substitution form is rejected.* Iso-WER contours of M_{sub} in $(\log T, \log d)$ at the median width, with the iso-latency line $8Td = \text{const}$. The fitted κ sits on its bound and $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$, so these contours are shown as a rejected model, not as a design rule.

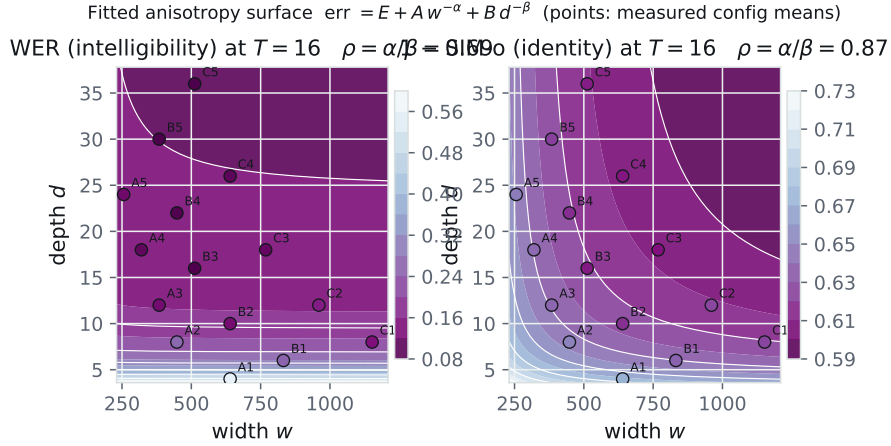


Figure 4: *Fitted anisotropy surface* at $T=16$ for both metrics, with the measured config means overlaid. The width coefficient A collapses to its lower bound for both metrics, which is why $\rho = \alpha/\beta$ is reported as unidentified in §4 even though the two-axis model beats the N -only model ($\Delta\text{AICc} = -15.5$ and -5.5).

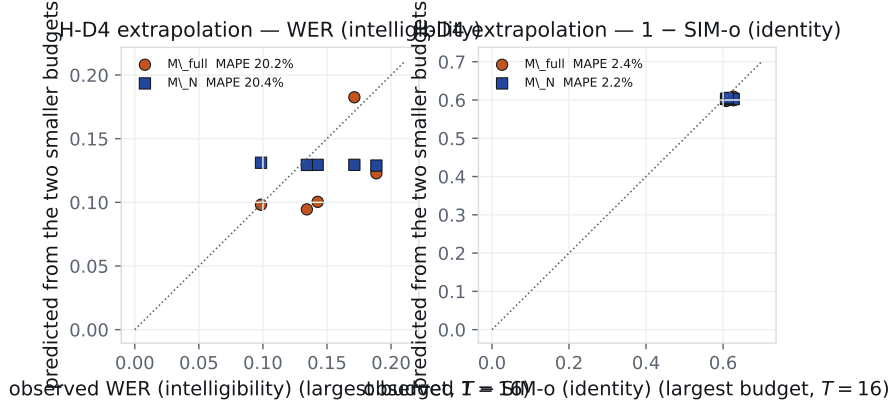


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4% vs 2.2%); WER does not (20.2% vs 20.4%).

205 C Fit diagnostics and exploratory analyses

206 **Parameters at bounds.** The width coefficient A reaches its lower bound (0) in the $T=16$ Part-A fit
 207 for both metrics, and κ reaches its upper bound in every one of the 2000 bootstrap replicates of M_{sub} ;
 208 both are reported as non-identification rather than as estimates. The irreducible term E reaches 0 for
 209 WER under M_{sep} .

210 **Weighting.** Points are weighted $1/\text{SE}^2$ with SE taken over seeds, floored at a quarter of the pooled
 211 seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored
 212 points is recorded in the released `fits.json`.

213 **Exploratory (not part of any declared claim).** An unweighted refit of both parts is released
 214 alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed.
 215 Both reproduce the sign and significance of $\Delta\tau$.

216 Use of Generative AI Disclosure

217 Generative AI was used as a coding assistant for the experimental harness and for language refinement
 218 of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis

219 plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and
220 take full responsibility for this paper.